JOHN WICK UNIVERSE DATASET
DECISION TREE EDA REPORT

Dataset Shape: (100000, 12)

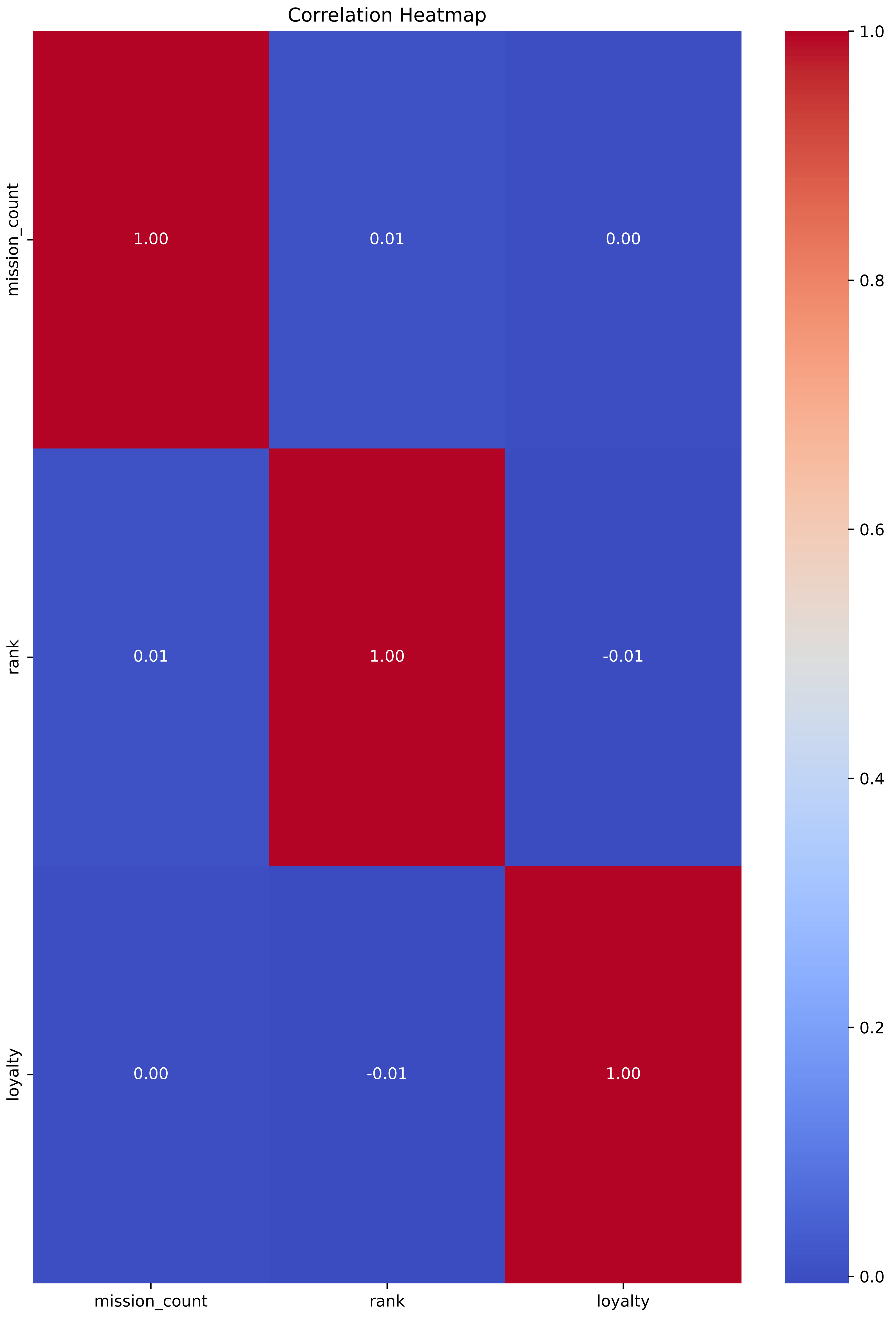
Target Variable: gender

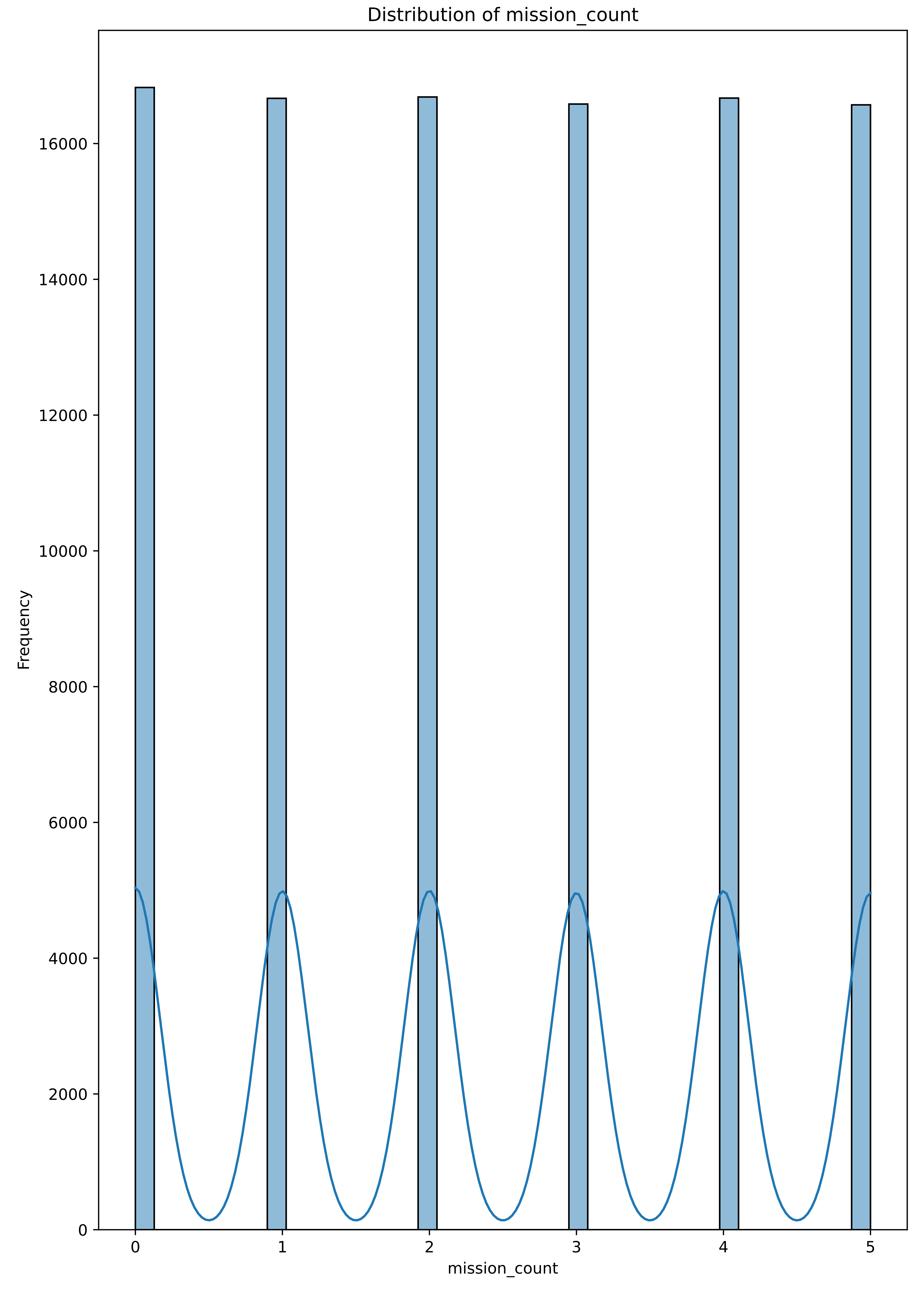
Numerical Features: 3
Categorical Features: 8

Missing Values: 0
Duplicate Rows: 4

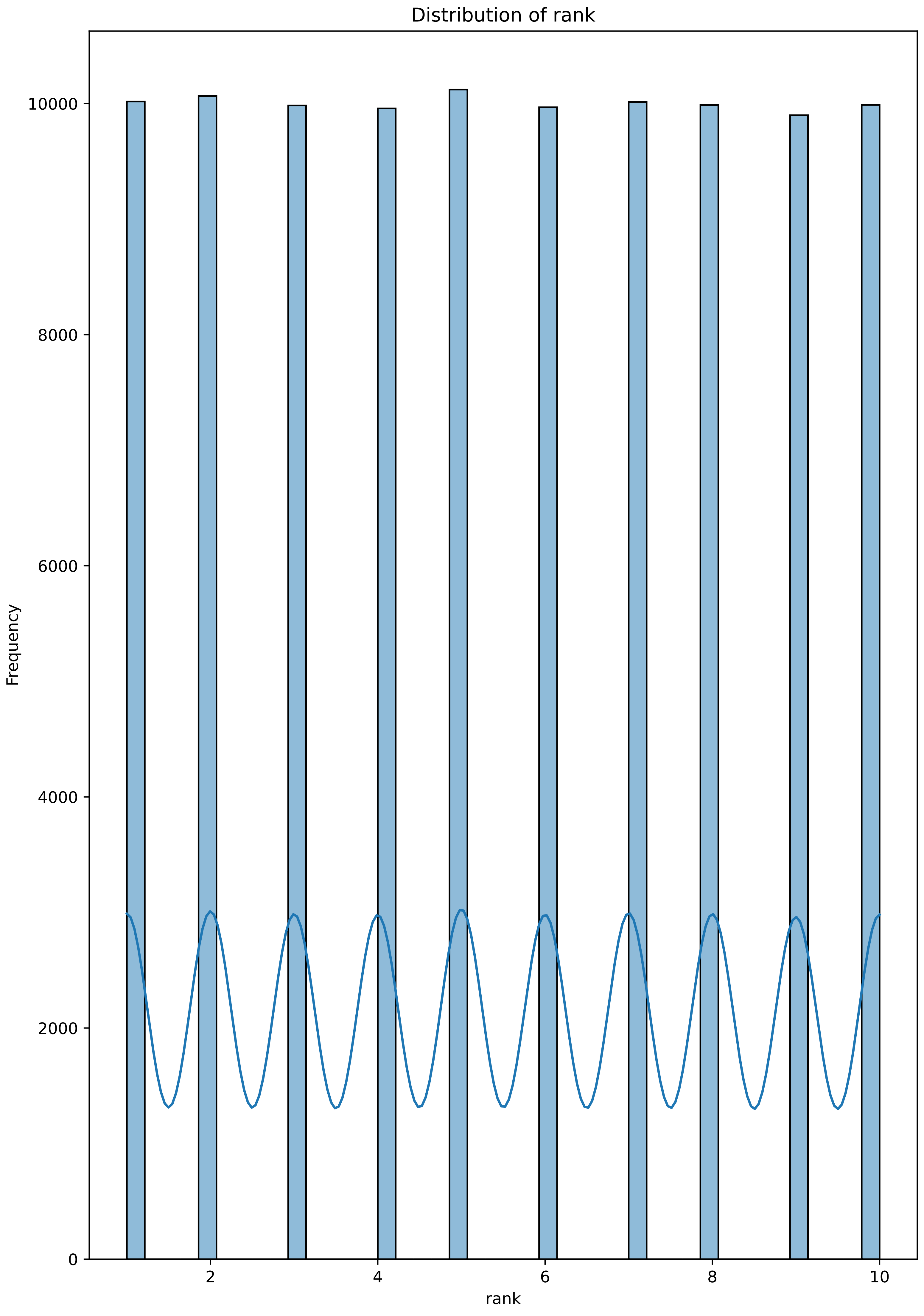
SPLITTING CRITERIA RESULTS

	Accuracy	Precision	Recall	F1 Score
Gini	1.0	1.0	1.0	1.0
Entropy	1.0	1.0	1.0	1.0

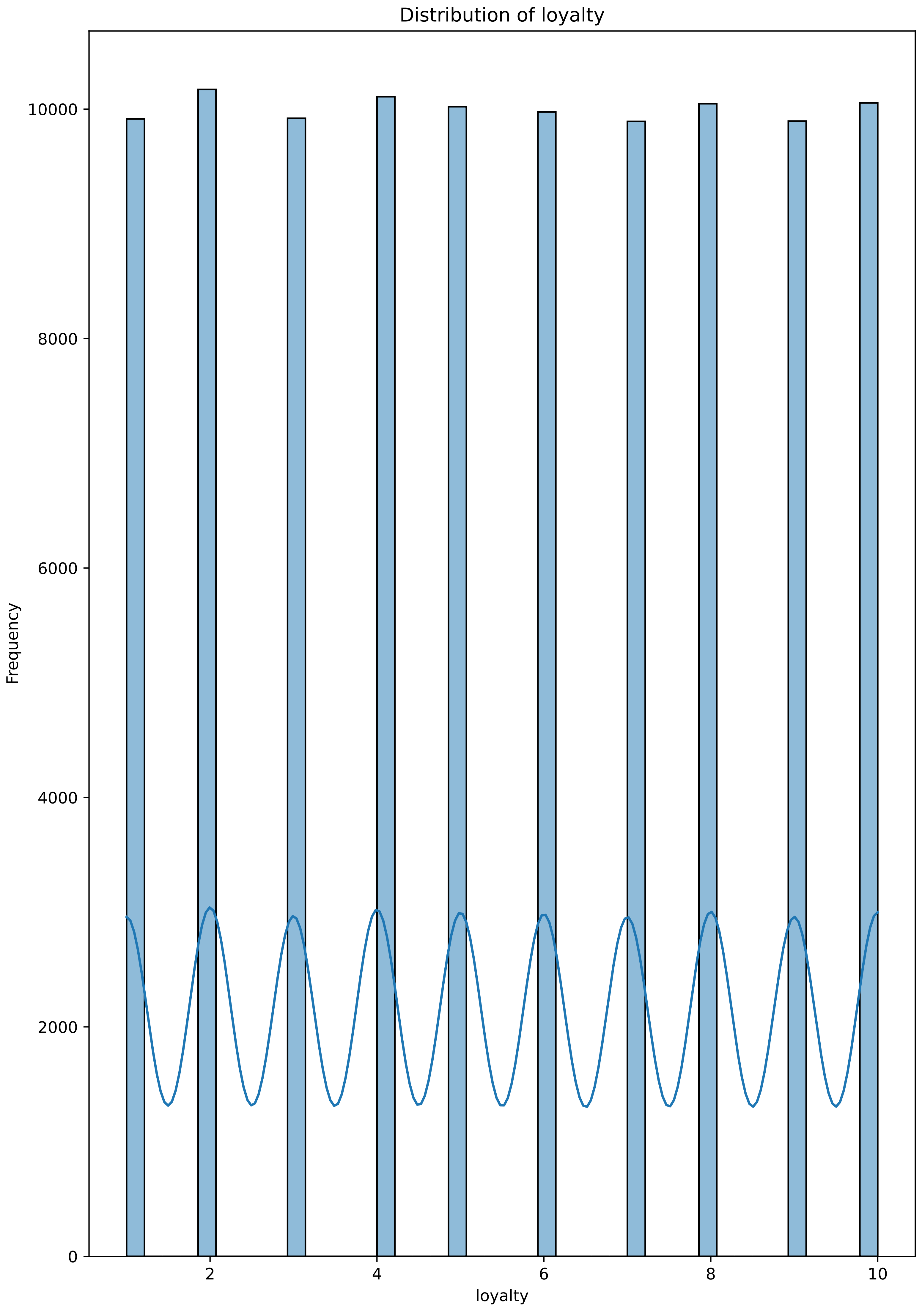

Correlation Heatmap
mission_count
rank
loyalty
1.00
0.01
0.00
0.01
1.00
-0.01
0.00
-0.01
1.00
mission_count
rank
loyalty
1.0
0.8
0.6
0.4
0.2
0.0

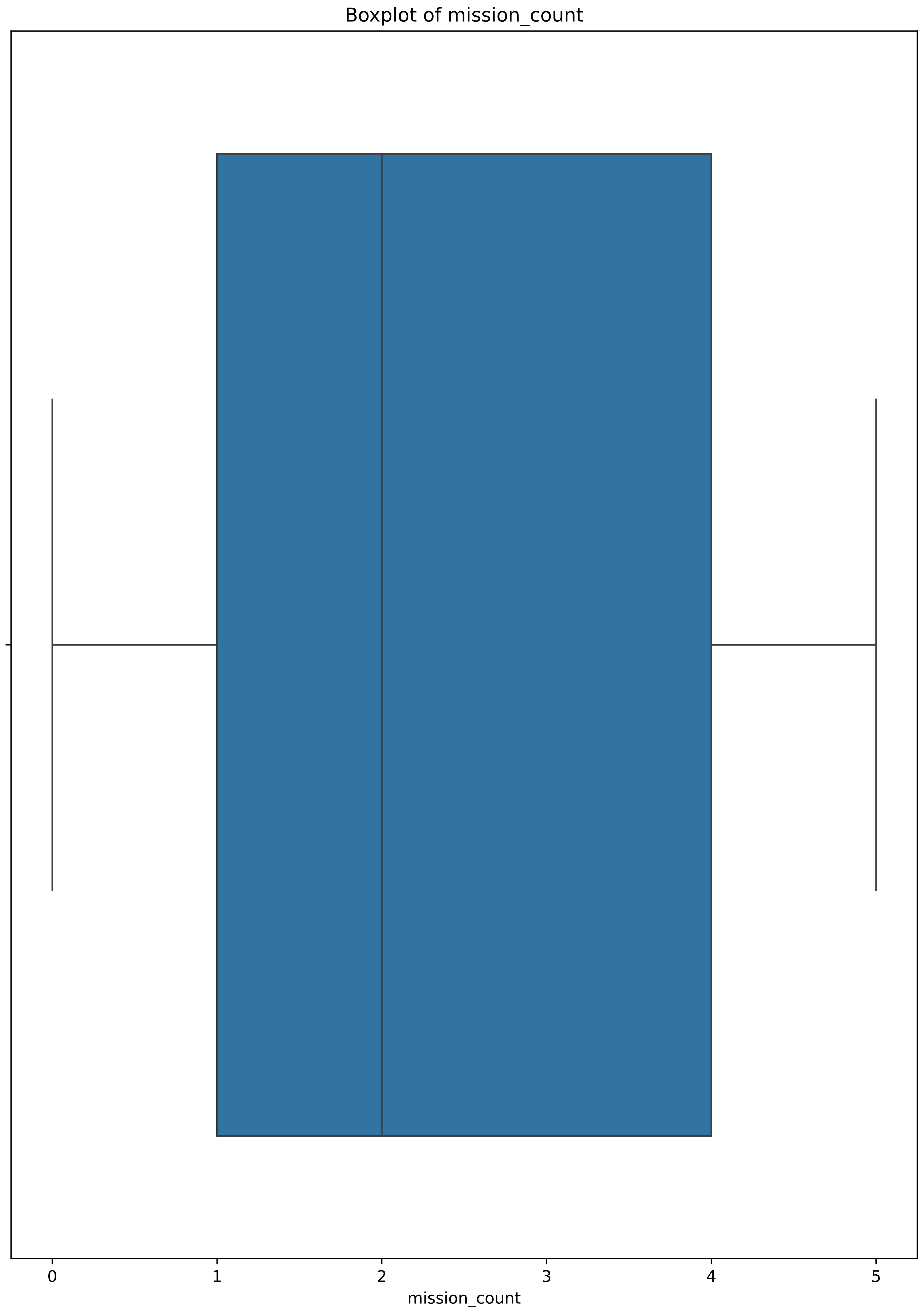


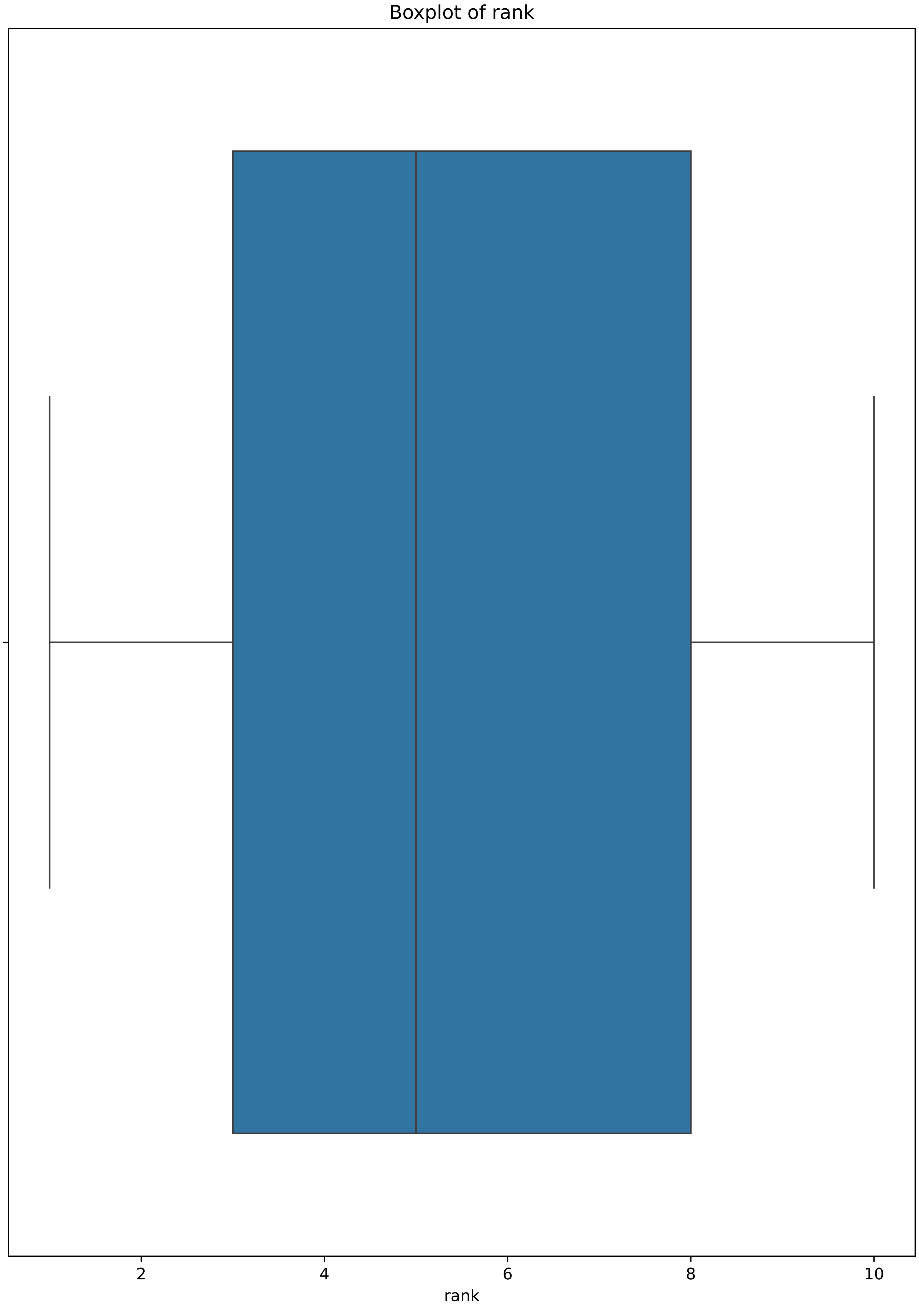
Distribution of mission_count
Frequency
mission_count
0
2000
4000
6000
8000
10000
12000
14000
16000
0
1
2
3
4
5



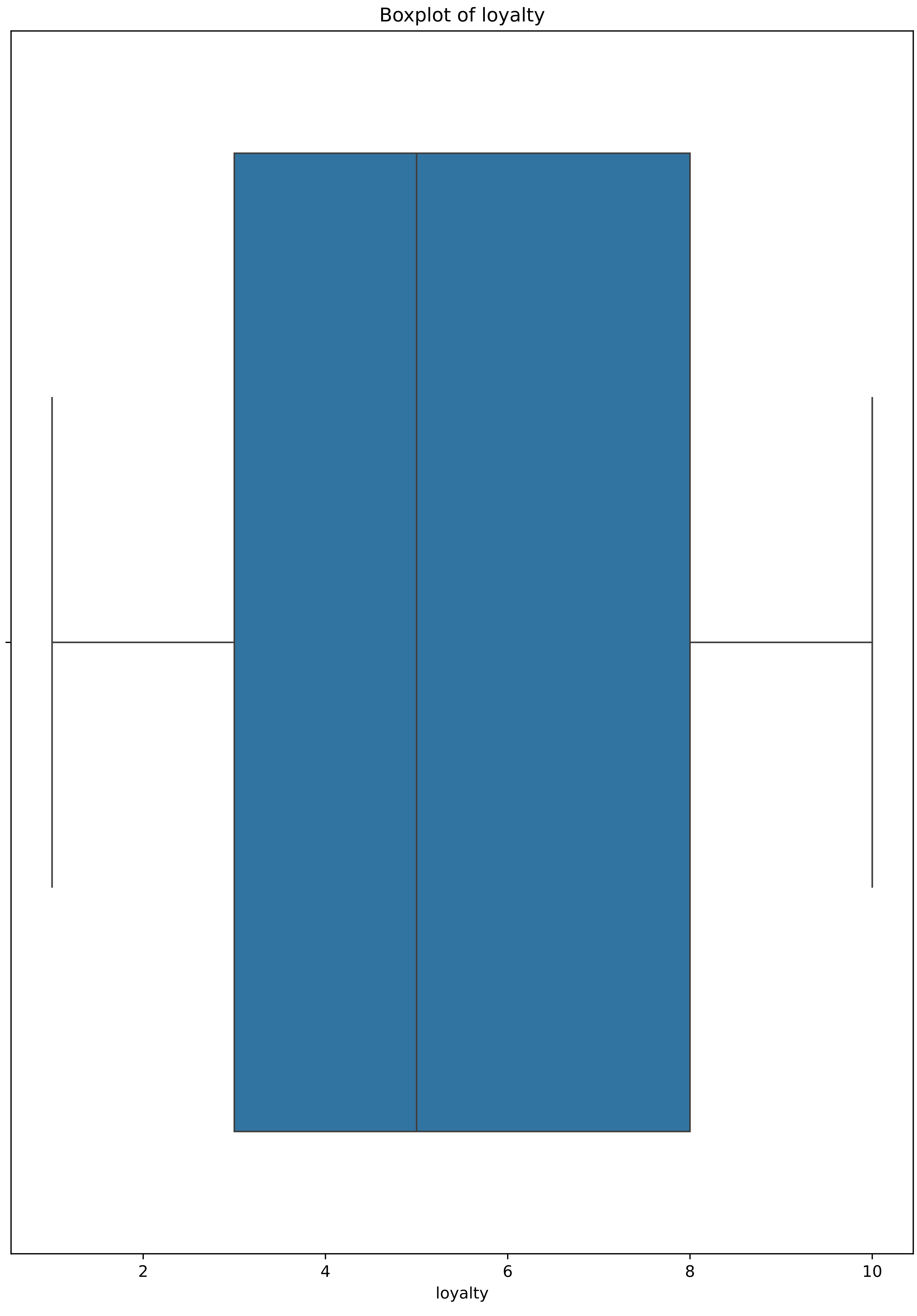
Distribution of rank
Frequency
rank
0
2000
4000
6000
8000
10000
2
4
6
8
10


Distribution of loyalty
Frequency
0
2000
4000
6000
8000
10000
2
4
6
8
10
loyalty

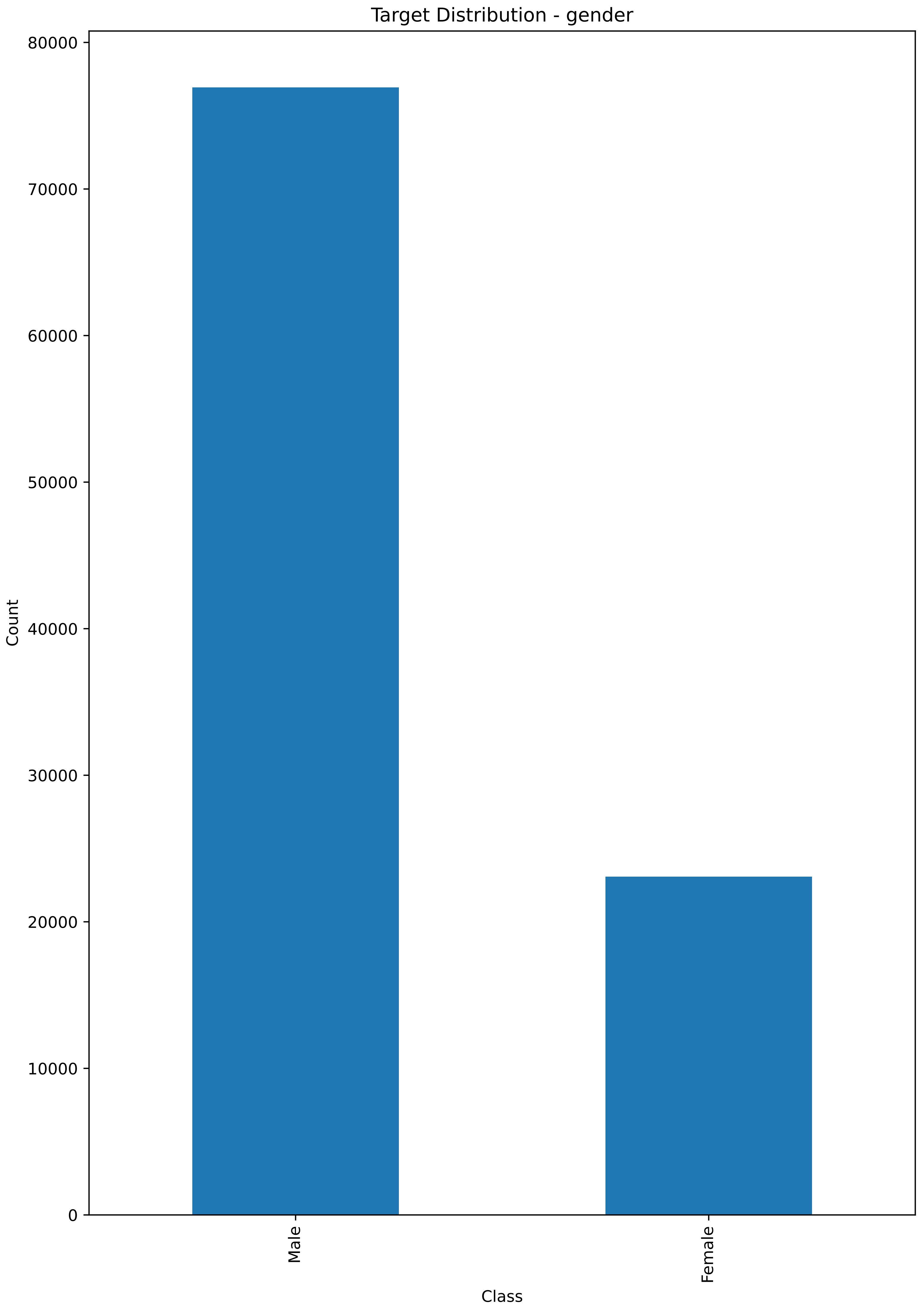

Boxplot of mission_count
0
1
2
3
4
5
mission_count



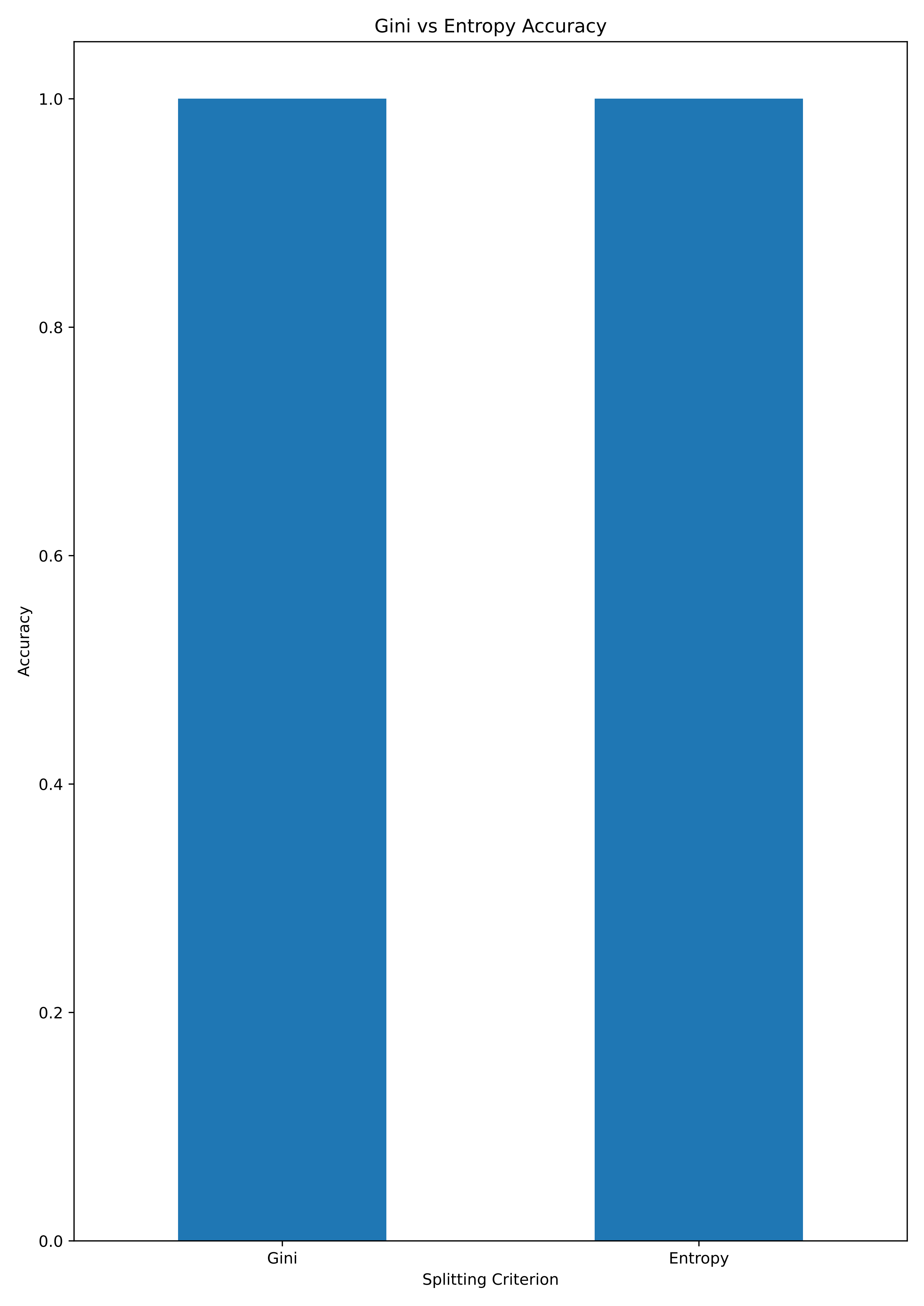
Boxplot of rank
2
4
6
8
10
rank

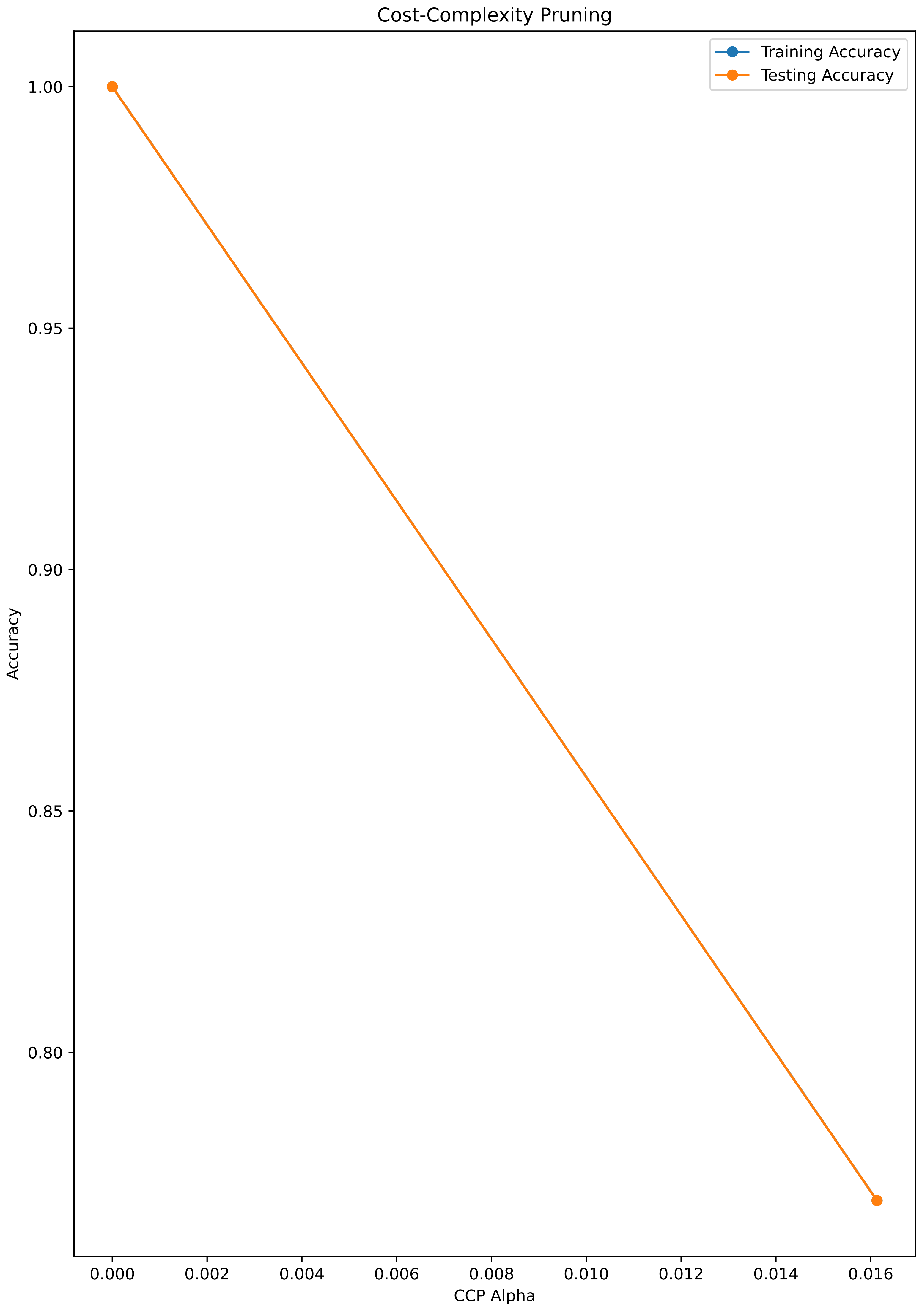


Boxplot of loyalty
2
4
6
8
10
loyalty

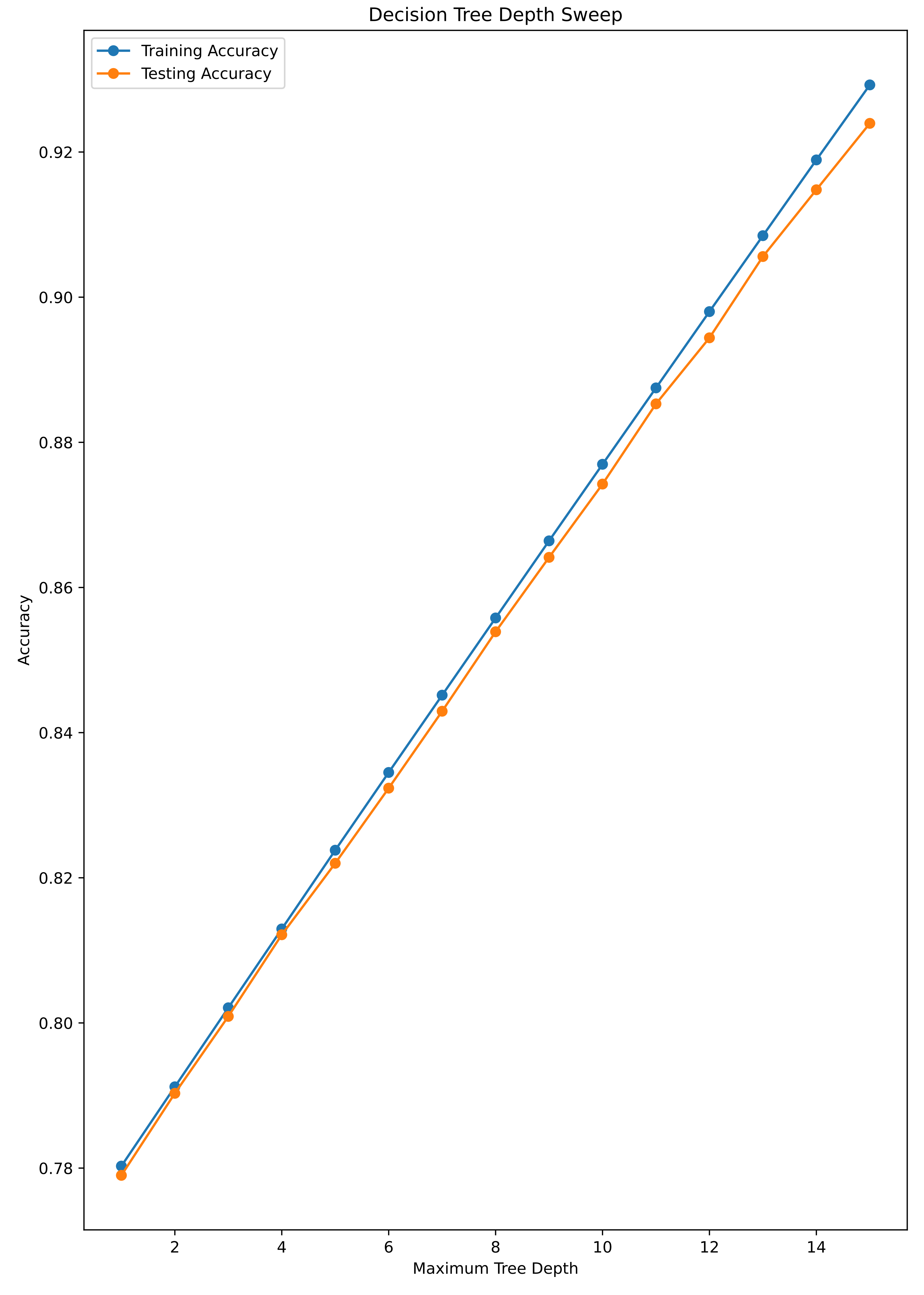


Target Distribution - gender
Count
80000
70000
60000
50000
40000
30000
20000
10000
0
Male
Female
Class

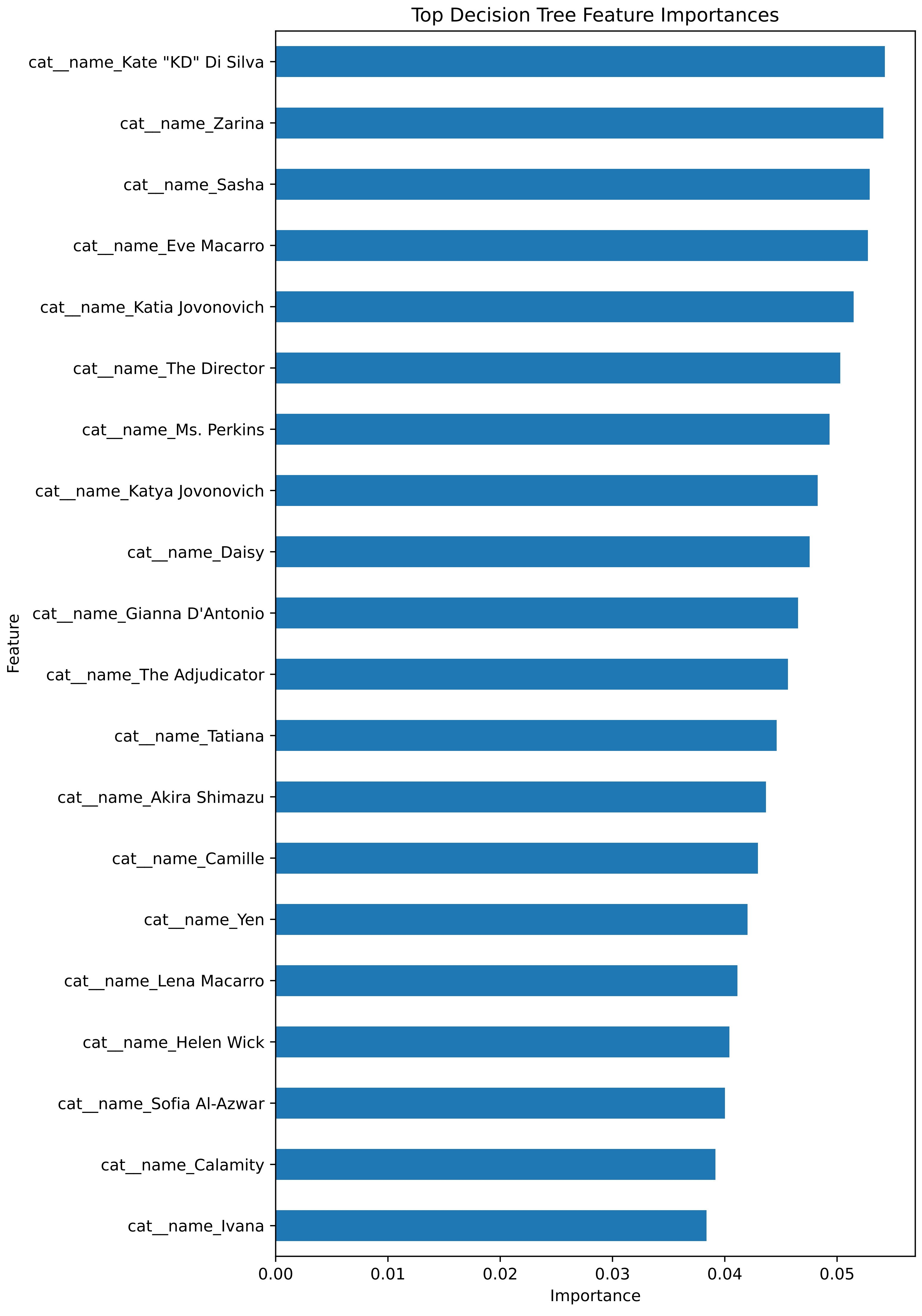

Gini vs Entropy Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
Gini
Entropy
Splitting Criterion



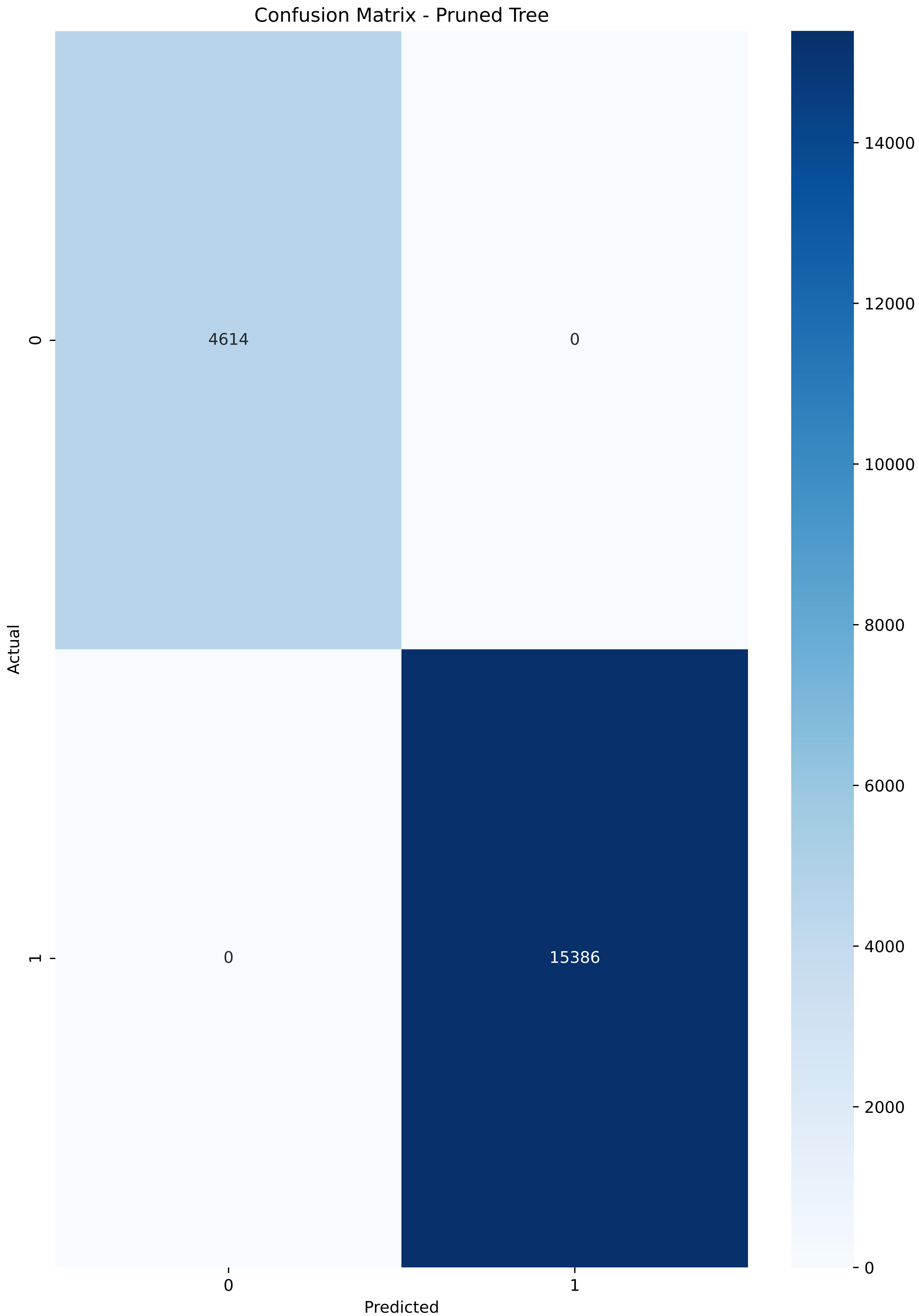
Cost-Complexity Pruning
Training Accuracy
Testing Accuracy
1.00
0.95
0.90
0.85
0.80
Accuracy
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
0.016
CCP Alpha



Decision Tree Depth Sweep
Training Accuracy
Testing Accuracy
0.92
0.90
0.88
0.86
0.84
0.82
0.80
0.78
Accuracy
2
4
6
8
10
12
14
Maximum Tree Depth



Top Decision Tree Feature Importances
cat__name_Kate "KD" Di Silva
cat__name_Zarina
cat__name_Sasha
cat__name_Eve Macarro
cat__name_Katia Jovonovich
cat__name_The Director
cat__name_Ms. Perkins
cat__name_Katya Jovonovich
cat__name_Daisy
cat__name_Gianna D'Antonio
cat__name_The Adjudicator
cat__name_Tatiana
cat__name_Akira Shimazu
cat__name_Camille
cat__name_Yen
cat__name_Lena Macarro
cat__name_Helen Wick
cat__name_Sofia Al-Azwar
cat__name_Calamity
cat__name_Ivana
Feature
0.00
0.01
0.02
0.03
0.04
0.05
Importance



Confusion Matrix - Pruned Tree
4614
0
0
15386
Actual
Predicted
0
1
0
1
14000
12000
10000
8000
6000
4000
2000
0

FINAL OBSERVATIONS

Target Variable: gender

Best Splitting Criterion by Accuracy: Gini

Best CCP Alpha: 0.000000

Pruned Tree Accuracy: 1.0000

Best Tree Depth: 15

Best Depth Testing Accuracy: 0.9240

Decision Tree classification was implemented using Gini Index and Entropy as splitting criteria.

Cost-Complexity Pruning was applied to reduce tree complexity and control overfitting.

A depth sweep was performed to study the effect of maximum tree depth on training and testing accuracy.